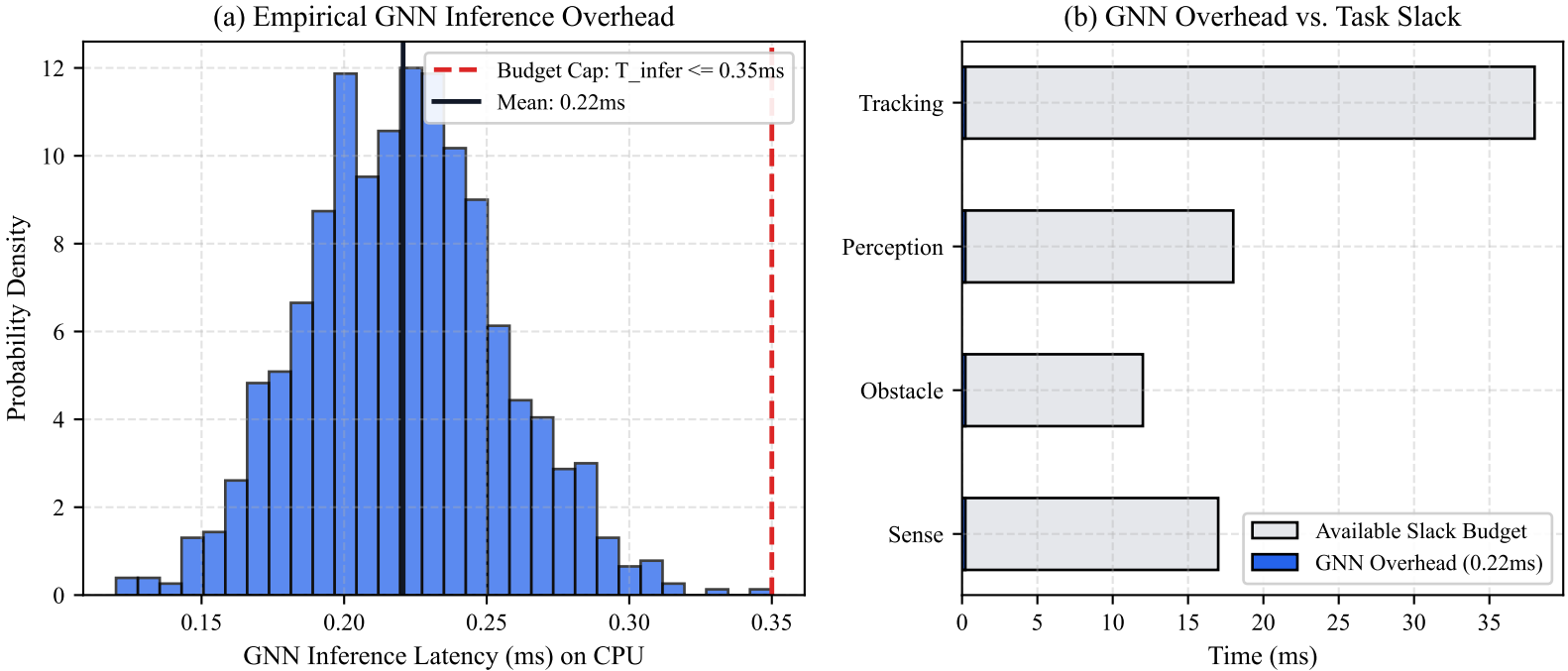
(a) Empirical GNN Inference Overhead
Budget Cap: T_infer <= 0.35ms
Mean: 0.22ms
Probability Density
GNN Inference Latency (ms) on CPU
0
2
4
6
8
10
12
0.15
0.20
0.25
0.30
0.35
(b) GNN Overhead vs. Task Slack
Tracking
Perception
Obstacle
Sense
Available Slack Budget
GNN Overhead (0.22ms)
Time (ms)
0
5
10
15
20
25
30
35